

High-Frequency Market Making via Ladder-Based Order Placement in Agent-Based Simulations

Joshua J Cable¹

¹ Department of Physics, University of Bath, Bath, BA2 7AY, United Kingdom

E-mail: jjc92@bath.ac.uk

Abstract. This paper details the development and benchmarking of a high-frequency market-making strategy implemented in the Bristol Stock Exchange, an agent-based simulation. The strategy employs a ladder-based system to dynamically build bid and ask price ladders around a smoothed mid-price drawn from real-time limit order book data. Liquidity signals help balance execution probability and profitability by minimising adverse-selection risk. Utilising a Poisson-driven order flow, the model optimises key parameters such as ladder depth and mid-price lag to achieve optimal performance. Extensive simulations statistically validate and demonstrate that this approach delivers robust, consistent, risk-adjusted returns compared to alternative strategies. Future research could incorporate realistic cost models, expand agent behaviour diversity to better mirror live markets, and integrate adaptive parameters to enhance responsiveness.

Prediction is very difficult, especially if it's about the future.

Niels Bohr

1. Introduction

In recent years, financial markets have become increasingly complex, driven by rapid technological advancements that have enabled the rise of high-frequency trading (HFT) and other algorithmic strategies [1]. Since the 1970s, the shift from in-person negotiation to fully electronic trading systems has allowed for greater precision and speed in execution [2]. As exchanges have evolved, so too have market participants, who now rely heavily on computational tools to manage trading decisions. HFT, in particular, exploits ultra-fast data processing and execution capabilities to capitalise on fleeting price discrepancies [3]. This amplifies the importance of cutting-edge technological infrastructure, while raising ongoing concerns about market fairness and systemic risk. According to a 2014 study by the U.S. Securities and Exchange Commission (SEC), HFT accounted for over half of all US equity trading volume [4], highlighting its significant influence. More recent industry estimates continue to place HFT's share between 50%-60% of total trade volume, reinforcing its role as a dominant force in modern markets. Moreover, HFT firms often act as market makers, with 63%-72% of total HFT volume on the Swedish stock exchange being from such activities [5].

Beyond strategies themselves, recent developments in market microstructure have seen a shift away from traditional quote-driven systems [6], in which designated market makers provided bid and ask quotes to facilitate trading. In modern limit order (LO) markets, any participant can post buy and sell orders directly, effectively acting as a market maker. This decentralised structure is implemented via a limit order book (LOB), essentially acting as a dynamic and real-time ledger that records all outstanding buy (bid) and sell (ask) LOs [7]. The most competitive buy and sell orders form the best bid and best ask, respectively, determining the prevailing market spread. The shape of the LOB, particularly the depth (number of orders at a given price) at each price level, reflects the current supply and demand across price points and offers crucial information on short-term liquidity and price formation dynamics [8].

While LO markets function efficiently under normal conditions, the flow of orders is not always stable. Sudden shifts, often termed jumps, can arise from macroeconomic announcements, geopolitical shocks, or endogenous feedback mechanisms [9]. For instance, US equity futures dropped sharply following the 2016 US presidential election, only to recover the next day in a rare reversal [10]. Similarly, the 2021 GameStop (GME) saga illustrated how social media fuelled buying pressure created a self-reinforcing cycle: as the stock rallied, market-makers hedging their outstanding call positions were forced to buy ever more shares, which in turn drove the price higher and produced extreme volatility [11]. Such

events underscore the need for trading strategies that can adapt to discontinuous price behaviour and highly dynamic market environments.

Advances in modelling these dynamics have focused on capturing the statistical properties of order arrivals, cancellations and executions using historical LOB data. A common modelling framework is to assume Poisson arrivals [12], where order events are both independent and memoryless. While this assumption may overlook temporal clustering or long-range dependence observed in real markets, it serves as a tractable baseline for linking liquidity, depth, and execution likelihood. Such assumptions are particularly prevalent in the modelling of interest rate derivatives and bond markets, where execution models often rely on simplified stochastic frameworks to capture order flow dynamics without introducing excessive computational complexity [13]. In these markets, the memoryless Poisson framework delivers tractable, closed-form solutions for execution risk, optimal order placement and price-impact modelling, combining analytical clarity with computational efficiency.

Despite this growing body of research, there is a notable gap in integrating LOB-driven execution models with adaptive, risk-aware strategies, especially within agent-based simulation frameworks that emulate microstructural market interactions. The Bristol Stock Exchange (BSE) [14] provides a useful platform to explore this integration, offering synthetic markets with customisable trader agents and real-time LOB data for a single, tradeable asset. This project proposes a novel adaptive order placement strategy that leverages LOB depth metrics and real-time liquidity conditions while accounting for execution risk under a Poisson-driven order flow regime. The strategy seeks to optimise the positioning of limit orders to balance execution probability and profitability, with specific attention to mitigating adverse selection. Implemented within the BSE, this model bridges the gap between microstructural signal processing and macro-level market responsiveness. The availability of high-frequency level II (two) data, defined as detailed order book information that includes all individual buy and sell orders, volumes, and their real-time updates [15], enables granular evaluation of trading strategy performance across varying liquidity regimes and market conditions.

2. The Bristol Stock Exchange

Imagine a busy marketplace where every buyer and seller writes their offers on a public board. In the Bristol Stock Exchange (BSE), this board is known as the Limit Order Book (LOB), a dynamic ledger that records all orders to buy or sell a particular asset. On one side, traders list their bids, indicating the price and number of shares (volume hereafter) they wish to purchase, while sellers post their asks, specifying the price and volume they are prepared to sell at [16]. Traders interact with the LOB by submitting limit orders (LO), which are instructions to trade at a designated price [17]. These orders are organised first by price, with the most competitive (highest bids and lowest asks) prioritised, and then by time, so that earlier orders take precedence when prices are equal. Each price level also displays the available volume, providing a clear picture of the depth of the market. Formally, the LOB at time t consists of discrete price levels, i , for bids, $B_i(t)$, and asks, $A_i(t)$, each associated with a volume $V_i(t)$. Figure 1 schematically illustrates this arrangement, highlighting the critical roles of the bid-ask spread and mid-price in driving trading decisions and reflecting the real-time balance of supply and demand.

2.1. BSE Market Participants

The functioning of the LOB is closely intertwined with the strategies of market participants [18]. In the BSE environment, a variety of trader archetypes are implemented to emulate realistic agent-based interactions. These range from simplistic strategies like Giveaway (GVWY), which simply mirrors the customer’s order price, to more adaptive strategies such as ZIP. Table 1 summarises the operational characteristics of these trader types. The diversity of agent behaviours within the BSE provides a controlled environment to study the impact of strategic heterogeneity on market dynamics. In the simulation, the customer’s order price is set by a predefined scheduling mechanism that mimics external supply and demand conditions. Specifically, the BSE’s `customer_orders()`

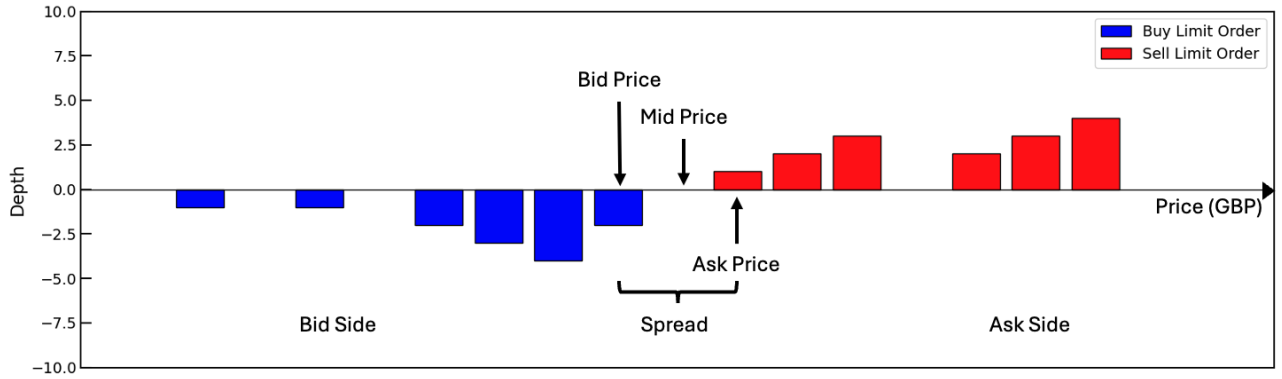


Figure 1: Schematic of a limit order book for a single security, showing buy (bids) limit orders (blue) and sell (asks) limit orders (red) at multiple price levels. The difference between the bid and ask forms the bid-ask spread, with the midpoint being the mid-price. Bar height represents the depth (cumulative volume) at each price level.

function generates limit prices for incoming buy and sell orders based on a predetermined schedule. These externally determined prices serve as a benchmark, with each trader receiving the prices and adjusting their quote, potentially incorporating price markups, markdowns, or learning-based modifications.

In real markets, customers, whether retail traders or institutional investors, initiate orders by specifying parameters such as price and volume. A broker or execution algorithm then works these orders in the market. While a naive broker might simply relay the order at the given price (as in the GVWY strategy), more sophisticated strategies attempt to optimise the execution by balancing urgency against market impact. Within the BSE framework, the customer's limit price provides the baseline, and the trader's performance is measured by its ability to secure a more favourable fill relative to that baseline. The notion of a loss occurs when execution worsens relative to the customer's limit price, buying above the maximum acceptable price or selling below the minimum acceptable price. Profit occurs when execution improves on the customer's limit price, buying below the maximum acceptable price or selling above the minimum acceptable price. This formulation ensures that profit and loss are calculated strictly from the trader's perspective.

2.2. Limit Order Book Liquidity and Trade Execution in the BSE

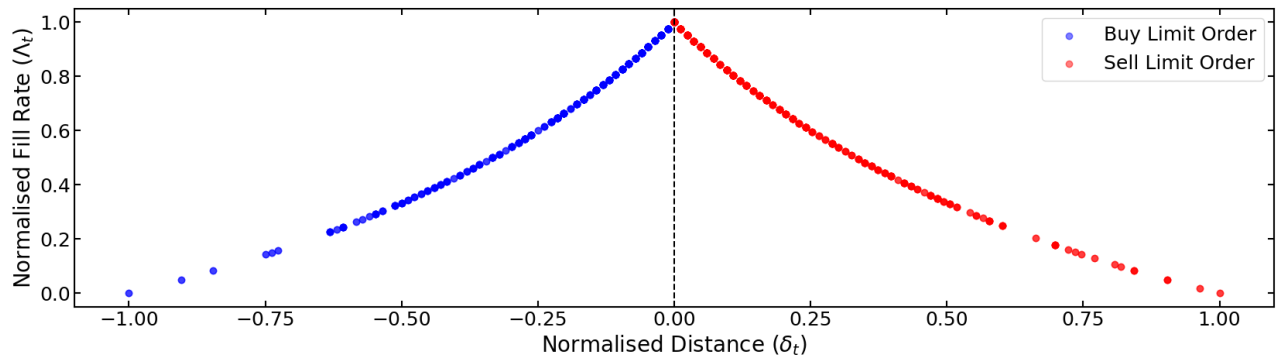


Figure 2: Fill rate as a function of δ , the order's distance from the mid-price, m_t . The x-axis represents the normalised distance ($\delta^\pm$), where negative values correspond to buy orders and positive values correspond to sell orders. The y-axis represents the fill rate (Λ_t), calculated as the executed quantity divided by the available depth. The dashed vertical line at $\delta = 0$ indicates the mid-price.

Market liquidity is the ease with which orders can be executed without significantly affecting price

Table 1: Overview of trader types used in the BSE, including their acronyms and characteristics.

Trader Type	Acronym	Characteristic Behavior
Giveaway	GVWY	The most basic trader, GVWY, simply replicates the customer's order price and immediately fills against the best opposing quote, making its profit solely from the bid-ask spread. It serves as a baseline, showing how a trading robot can operate with minimal decision-making [14].
Zero-Intelligence Constrained	ZIC	ZIC introduces randomness into its pricing. ZIC selects bid or ask prices at random, but within strict boundaries that prevent it from quoting prices that would result in a loss. This constrained randomness shows that even without complex intelligence, a trading algorithm can still generate viable market activity by ensuring that every trade has a positive profit potential [19].
Shaver	SHVR	Shaver actively monitors the LOB and slightly undercuts the best available prices. By shaving a small amount, SHVR aims to be the most attractive counter-offer in the market while still respecting the customer's specified price limit. This approach highlights how small, incremental improvements can have a significant impact in competitive trading environments [14].
Sniper	SNPR	Sniper maintains a cautious posture for most of the trading session, only becoming aggressive as the market session nears its close. With a built-in sense of urgency, it rapidly increases its price improvements when time is short, thereby "stealing" trades that might otherwise be lost if it continued with a conservative approach. This time-sensitive tactic underscores the importance of timing in trading strategies [14].
Zero-Intelligence Plus	ZIP	ZIP is the most sophisticated approach among BSE agents. ZIP uses adaptive learning techniques to adjust its profit margin dynamically. It starts with a quote based on a simple margin over the customer's limit price but continuously refines its pricing strategy through machine learning-inspired adjustments. This enables ZIP to converge towards the market equilibrium while managing its profit expectations more effectively [20].

[21]. In the BSE, liquidity is quantified by the depth of the LOB, which is defined as the total volume of orders available at each price level at a given time [22]. At any given time, the total depth can be written as a sum over all price levels, i ,

$$\mathcal{D}_t = \sum_i V_i(t). \quad (1)$$

This measure is critical because it influences the probability that a newly submitted order will be executed. Central to this is the mid-price, m_t , calculated as

$$m_t = \frac{B_{\max}(t) + A_{\min}(t)}{2}, \quad (2)$$

where $B_{\max}(t)$ and $A_{\min}(t)$ denote the best bid and ask, respectively. The mid-price acts as an estimate of fair market value and is often used as an anchor in trading strategies [23]. As new orders arrive and existing orders are executed, both the LOB depth and the mid-price fluctuate continuously.

Table 2: Table demonstrating decision outcomes, determined by the Limit Order Book liquidity.

Distance, δ (Normalised)	Fill Rate, Λ_t	Trading Decision
Near Spread ($\delta \approx 0$)	High	Market Order (Aggressive)
Shallow Depth ($-0.2 < \delta < 0.2$)	Medium	Limit Order Near Best Bid/Ask
Deep in LOB ($\delta < -0.5$ or $\delta > 0.5$)	Low	Avoid Trading

It is conventional to model the execution probability of a limit order as an exponential function of its price offset from the mid-price [24],

$$h^\pm(\delta; \kappa_t) = e^{-\kappa_t^\pm \delta}, \quad (3)$$

where δ is the order's price distance from the mid-price and κ_t is a liquidity-dependent decay parameter [25]. In more liquid scenarios, κ_t is larger, favouring shallower order placement near the mid-price; conversely, smaller values in illiquid conditions imply deeper placements, mitigating adverse selection risk. The execution probability per unit time, or fill rate, is then defined as

$$\Lambda_t^\pm = \lambda_t^\pm h^\pm(\delta; \kappa_t) \quad (4)$$

where $\lambda_t^\pm$ is the arrival rate of customer orders. Figure 2 visualises this relationship from BSE data on how fill rates vary with order placement distance from the mid-price, demonstrating that for either bids or asks, the rate is the same at an equal distance. The fill probability $\Lambda_t^\pm$ not only depends on the inherent order arrival rate, but also on the distance δ from the mid-price. The BSE implements a continuous double auction with strict price-time priority: incoming trader-generated bids and asks are matched directly against the resting limit order book. As a result, execution probabilities emerge purely from the dynamics of order flow and book depth, transparently capturing both execution risk and adverse selection in a fully deterministic framework.

The optimal order placement distance $\delta^\pm$ from a strategic perspective is determined by maximising the expected gain, defined as the distance δ multiplied by the probability that an order placed at that distance will be executed,

$$\delta^\pm = \max_{\delta} \delta \lambda_t^\pm h^\pm(\delta; \kappa_t). \quad (5)$$

This formulation highlights that order placement strategies should dynamically adapt to prevailing liquidity conditions. Table 2 outlines guidance on optimal actions based on price distance and execution probability, helping traders decide whether to submit aggressive market orders, passive limit orders, or to refrain from trading.

Order Flow Imbalance (OFI),

$$\text{OFI} = \frac{V_{\text{bid}} - V_{\text{ask}}}{V_{\text{bid}} + V_{\text{ask}}}, \quad (6)$$

captures the net buying or selling pressure. The correlation of OFI with mid-price, m_t , and transaction price, P_t can signal whether strong imbalances push the market upward or downward. Transaction price typically clusters near the mid-price but can deviate with volatility or abrupt shifts in order flow [26]. Taken together, these relationships underscore the intertwined dynamics of a complex, many-trader system, where local changes can propagate quickly through the LOB, affecting overall market liquidity and price formation. Such insights are vital in understanding the microstructure of financial markets.

To examine LOB variables, Figure 3 shows a corner plot of distributions and pairwise dependencies between best bid, ask, mid-price, transaction price, and order flow imbalance. The data is from a baseline simulation, appendix 8.2, of 1000 market sessions (each 120 seconds (s) long) with drip-Poisson customer order arrivals and a 0.1 s order interval. Despite the simulated nature, this setup reproduces several stylised facts of real-world LOB markets, such as high-frequency price formation,

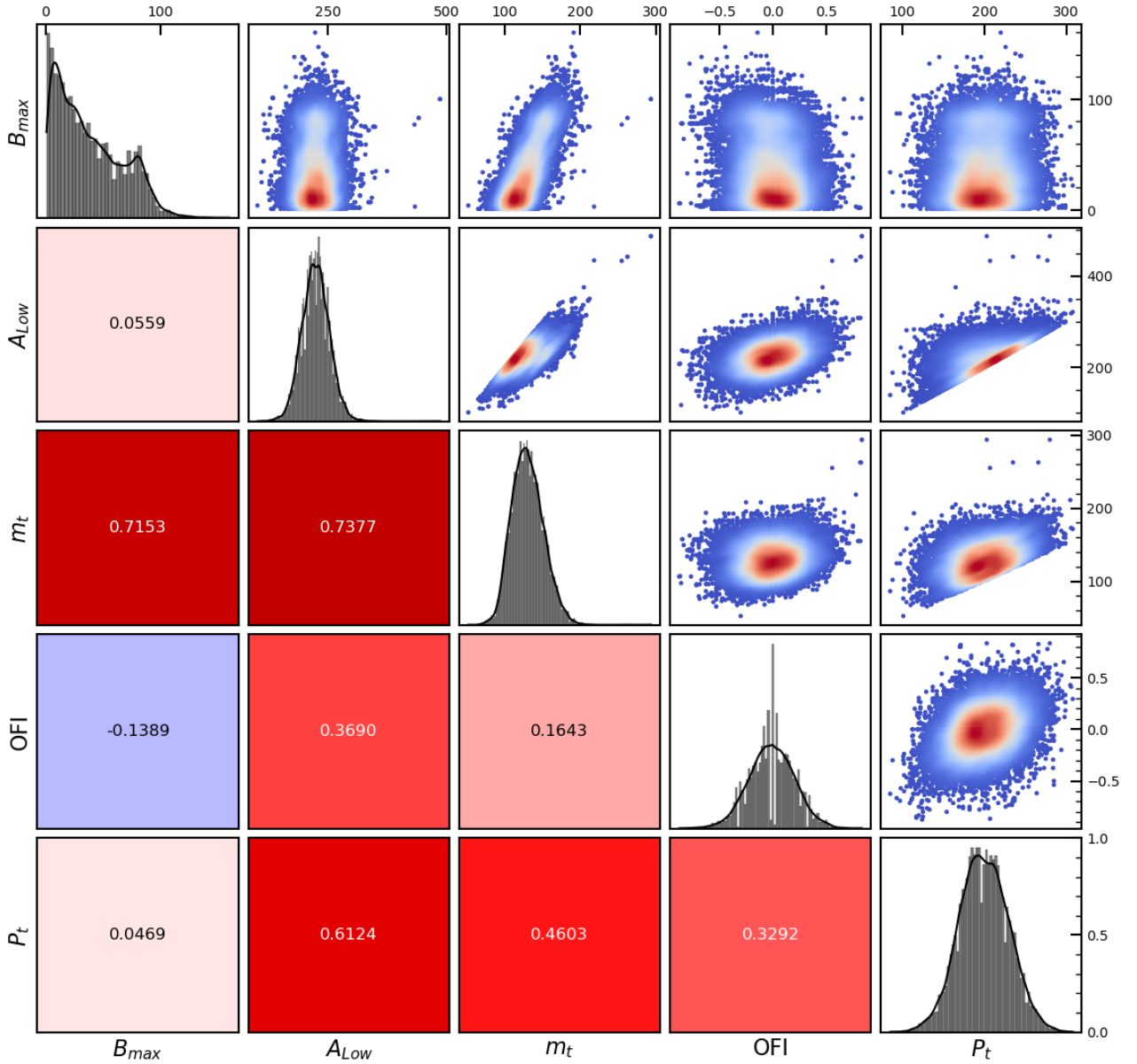


Figure 3: Pairwise relationships among the best bid, best ask, mid-price, order flow imbalance, and transaction price, generated from the baseline simulation setup. Diagonal panels display univariate distributions (histograms with kernel density estimates), while off-diagonal panels reveal bivariate relationships. In the lower triangle, the Pearson correlation coefficient is computed and visually represented via a diverging background colour, whereas the upper triangle shows a density scatter plot.

heavy-tailed order flow, and transient volatility, documented in prior studies [27]. As a result, the emergent correlations and distributions are broadly representative of typical LOB dynamics. Along the diagonal of the plot, histograms and kernel density plots highlight whether each variable is symmetric, skewed, or heavy-tailed patterns suggestive of non-Gaussian or even critical dynamics. Off-diagonal subplots depict pairwise correlations, revealing how certain variables move together or in opposition.

Analysing these interactions shows that m_t is very tightly linked to both $B_{\max}$ (corr = 0.72) and A_{Low} (corr = 0.74), while OFI has a weaker, negative relationship with $B_{\max}$ (corr = -0.14) but a

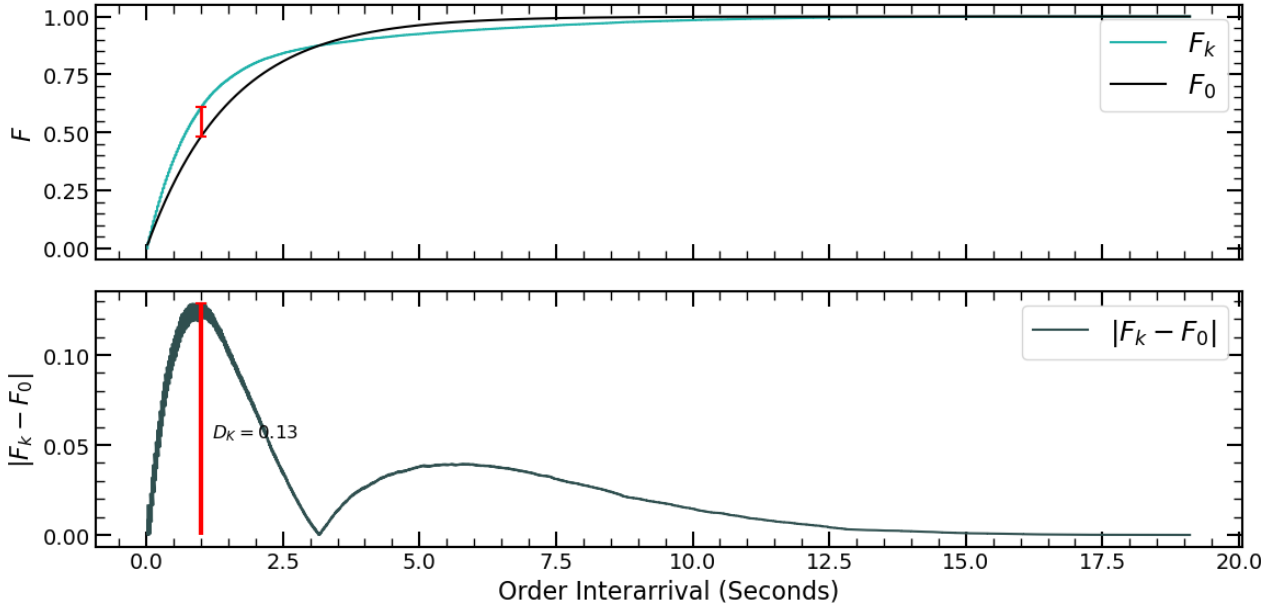


Figure 4: Example of the Kolmogorov-Smirnov (KS) test performed on the baseline simulation setup. The KS test statistic D_K is the largest value of the absolute difference between the empirical distribution function F_K and the reference distribution F_0 . The p-value was found to be 0.23.

modest positive link to A_{low} ($\text{corr} = 0.37$). Since m_t tracks the market's consensus value [28], quoting relative to a smoothed mid-price captures the central tendency of executions across liquidity regimes, making it an ideal anchor for adaptive market-making. Monitoring deviations from m_t lets traders identify when to lean into the market (by tightening quotes) or pull back (by widening them), thereby balancing aggressiveness against adverse-selection risk.

2.3. BSE's Memoryless Poisson Framework

Order arrivals in the BSE are structured such that they follow a time-homogeneous process, referred to as an independent pure Poisson process, $N(\tau)$ [29]. This approach treats each order as a discrete event that occurs randomly over time, following a continuous point process framework [30]. More specifically, the change in the number of orders over an infinitesimal time interval is given by

$$dN(\tau) = d\tau \sum_k \delta_1(\tau - \tau_k), \quad (7)$$

where δ_1 is the inclusive Dirac delta function, appendix 8.1, and τ_k denotes the arrival time of the k -th order. This point process can be characterised by an intensity function,

$$\lambda(\tau) \equiv \mathbb{E} \left[\lim_{\Delta\tau \rightarrow 0} \frac{N(\tau + \Delta\tau) - N(\tau)}{\Delta\tau} \right] \equiv \mathbb{E} \left[\frac{dN(\tau)}{d\tau} \right]. \quad (8)$$

The memoryless Poisson framework guarantees analytical tractability while still reproducing the high-frequency order-flow patterns typical of interest-rate and bond markets.

The validity of the Poisson structure is tested via the Kolmogorov-Smirnov (KS) test, particularly crucial in a market with many active traders, as the interactions among diverse agents can introduce clustering and dependencies that may cause deviations from a pure Poisson process. The KS test compares the empirical cumulative distribution (empirical) of order interarrival times with the expected exponential distribution (reference). Consistency between the two supports the randomness and independence assumptions. In this test, the empirical distribution function $F_K(x)$ constructed

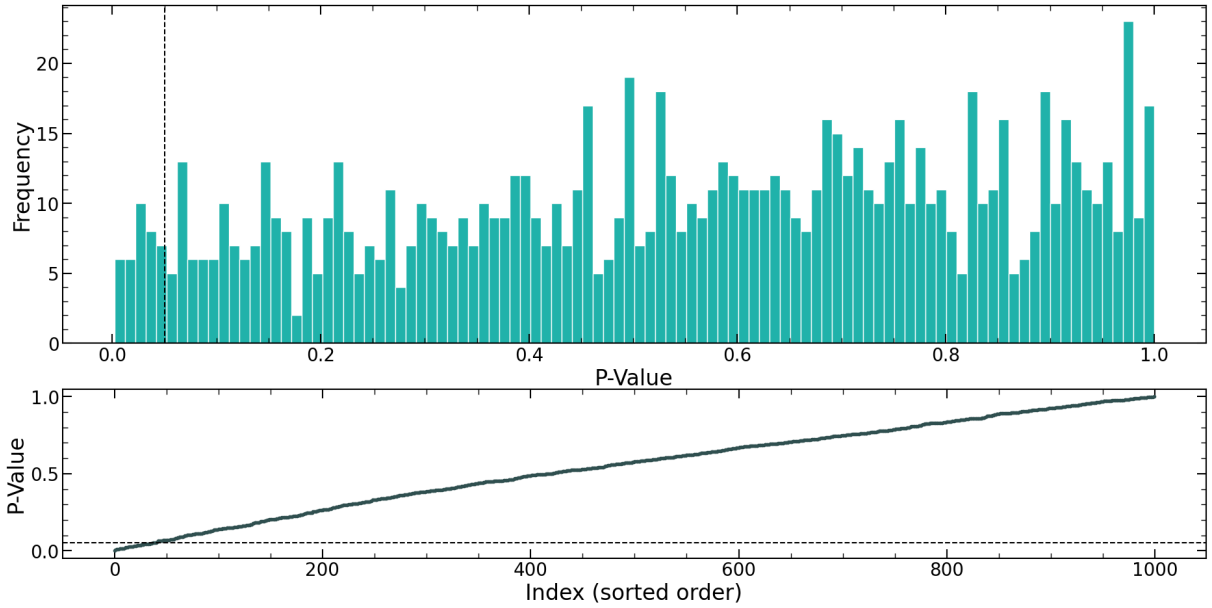


Figure 5: Distribution and sorted values of p-values obtained from Ljung–Box tests on order interarrival times across the baseline simulation. The top panel shows a histogram of the p-values, with a dashed line at 0.05 indicating the usual significance threshold. The bottom panel plots the same p-values in ascending order, again highlighting the 0.05 level. The predominance of p-values above 0.05 suggests that the null hypothesis of no autocorrelation cannot be rejected for most sessions, reinforcing the assumption of memoryless arrivals.

from $K = 1000$ independent observations is compared against a reference distribution $F_0(x)$. The empirical distribution is defined as

$$F_K(x) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}_{\{X_k \leq x\}}, \quad (9)$$

where $\mathbf{1}_{\{X_k \leq x\}}$ equals 1 if $X_k \leq x$ and 0 otherwise. The KS statistic is given by

$$D_K = \sup_x |F_K(x) - F_0(x)|, \quad (10)$$

where $\sup_x$ denotes the supremum (maximum) over all x , so that D_K is simply the largest absolute difference between the empirical and reference distributions. Under the null hypothesis $H_0 : F_K = F_0$, the statistic $\sqrt{K}D_K$ converges to a Kolmogorov distribution, independent of F_0 . Figure 4 illustrates a KS test applied to the baseline simulation setup, yielding a p-value of 0.23 and confirming the Poisson nature of the BSE order interarrival times.

Verifying that order arrivals in the BSE follow a Poisson process is essential for designing effective trading strategies and ensuring model validity. The assumption of a Poisson process, where orders arrive independently and randomly over time, without memory or systematic bias enables simplification in the modelling of market behaviour. This is particularly prevalent in areas such as adverse selection, execution risk, and liquidity forecasting [31]. Through ruling out hidden patterns or correlations in order flow, traders can rely on cleaner, more tractable models that are both analytically robust and computationally efficient [32].

To confirm the lack of hidden temporal patterns or memory effects, the Ljung–Box test (LB) is used to evaluate autocorrelation of order inter-arrival times [33]. The test statistic is derived from normalised sample correlations, $\hat{\rho}_h$, at various lags, which ensures any subtle dependencies are detected. With

the normalised sample auto-correlation at lag h defined as

$$\hat{\rho}_h = \rho_h[X_{0:k}] = \frac{\text{COV}(X_{h:k}, X_{0:k})}{\text{SD}[X_{h:k}] \cdot \text{SD}[X_{0:k}]}. \quad (11)$$

The LB statistic can be formed such that

$$Q = K(K+2) \sum_{h=1}^H \frac{\hat{\rho}_h^2}{K-h}, \quad (12)$$

which, under the null hypothesis of no autocorrelation, is approximately chi-squared distributed. Figure 5 demonstrates that across 1000 simulations, order flows exhibit negligible autocorrelation, confirming the Poisson nature. The subplot demonstrates that fewer than 5% of the simulation p-values lie below 0.05, further indicating no significant correlation.

Together, these validations provide a robust foundation for linking micro-level LOB dynamics to macro-level flow behaviour, and thereby support the investigation into execution models and their sensitivity to market microstructure variables.

3. A Market Maker Strategy

3.1. Design

Leveraging the statistical insights on order flow and LOB dynamics above, a market-making (MM) strategy was developed for the BSE framework. The primary goal was to supply liquidity by consistently posting competitive bid and ask orders while maintaining profitability through controlled inventory exposure and minimised adverse selection.

At the core of the strategy lies the concept of constructing a price ladder on both sides of the LOB. Using the best bid ($B_{\max}$) and best ask (A_{low}) as reference anchors, the algorithm dynamically calculates a smoothed mid-price, m_{eff} , reducing the sensitivity to transient price shocks. This is accomplished by retaining a rolling history of mid-prices and choosing a lagged value if sufficient data is available,

$$m_{\text{eff}} = \begin{cases} m_{t-j} & \text{if } t \geq j \\ m_t & \text{otherwise} \end{cases} \quad (13)$$

where j is a configurable parameter for the amount of lag. This effective mid-price serves as a fair-value benchmark that drives both the buy-side and sell-side ladders.

The MM strategy builds discrete price ladders of total quantity L , for both buying and selling so that any filled order remains profitable relative to the mid-price. The minimum allowable price increment is defined by the tick size [30], δ_{tick} (defaulting to one in the BSE), but can be reduced for more aggressive trading. A profit margin, π , is also enforced within the MM strategy to ensure that a minimum spread is maintained, mitigating the chances of trades resulting in a loss.

Buy Ladder:

For the i th level on the bid side (with $i = 0, 1, \dots, L-1$), the initial desired buy price is calculated as

$$p_{\text{buy}} = B_{\max} + \delta_{\text{tick}} - i\delta_{\text{tick}}.$$

To guarantee profitability, this price is clipped such that $p_{\text{buy}} \leq m_{\text{eff}} - \pi$. Thus, the final price for level i is defined as

$$p_{\text{buy},i} = \min \left\{ B_{\max} + \delta_{\text{tick}} - i\delta_{\text{tick}}, m_{\text{eff}} - \pi \right\}. \quad (14)$$

Sell Ladder:

Similarly, for the i th level on the ask side, the initial desired sell price is

$$p_{\text{sell}} = A_{\text{low}} - \delta_{\text{tick}} + i\delta_{\text{tick}}.$$

This price is clipped from below to ensure $p_{\text{sell}} \geq m_{\text{eff}} + \pi$. Thus, the final sell price is determined as:

$$p_{\text{sell},i} = \max \left\{ A_{\text{low}} - \delta_{\text{tick}} + i\delta_{\text{tick}}, m_{\text{eff}} + \pi \right\}. \quad (15)$$

This design ensures each order placed remains theoretically profitable if filled while minimising volatility-induced crossing of orders. An inventory control mechanism bounds the MM's position to limit directional exposure. Should the computed ladders overlap, that is, $\max p_{\text{buy},i} \geq \min p_{\text{sell},i}$, the strategy reverts to using the effective mid price, m_{eff} , to reset

$$p_{\text{buy},i} = m_{\text{eff}} - \delta_{\text{tick}} - i\delta_{\text{tick}}, \quad p_{\text{sell},i} = m_{\text{eff}} + \delta_{\text{tick}} + i\delta_{\text{tick}}, \quad (16)$$

maintaining a sufficient spread between quotes.

Although this framework is conceptually aligned with models that link fill probabilities to distance from the mid-price, as suggested by the exponential decay functions, the current implementation does not explicitly compute those fill rates. Instead, it relies on fixed margins, π , and a smoothed mid-price anchor to determine order placement and profitability. This results in the MM strategy effectively always having orders at each of the levels in Table 2.

The reliance on a smoothed mid-price is particularly effective because it represents the market's consensus value, reducing short-term noise. By placing staggered limit orders around a smoothed mid-price, the MM strategy captures spread profits, reduces exposure to unfavourable prices, and improves the chances of achieving risk-adjusted returns. The confirmation of independent Poisson order arrivals allows the strategy to conclude that past flows do not predict future arrivals. This independence justifies simpler, more robust models for execution probability forecasting and ladder parameter calibration. Collectively, the methodological choices allow the MM strategy to consistently provide liquidity and extract profits in HFT Poisson environments.

3.2. Strategy Optimisation

While the ladder-based quoting framework is structurally sound, the overall profitability is dependent on parameter selection. Two hyperparameters are of primary importance, the number of ladder levels, L , and the lag, j , used in computing m_{eff} .

To investigate their impact, a grid search was conducted over 1000 baseline BSE simulations, varying values of L and j , with all traders from Table 1 present. The results, visualised in Figure 6, reveal a non-linear sensitivity to the parameters. Each cell in the heatmap corresponds to a specific (L, j) pair, with lighter or darker shading indicating better or worse average profits, respectively. It can be seen that increasing L enhances market coverage and potential profit, but also increases exposure to adverse market movements. Conversely, adjusting j changes the strategy's responsiveness to microstructural fluctuations: a short lag tracks price changes closely but may overreact to noise, while a long lag smooths volatility at the cost of delayed reaction. The trade-off can be summarised by four key failure modes:

- If L is too low, the market maker places too few orders, capturing less of the market's available opportunities.
- If L is too high, the trader may overexpose itself across many price levels, risking substantial losses if the market moves quickly in one direction.
- If the j is too short, the effective mid-price can become noisy, leading to frequent re-quoting and potential overreaction to small price fluctuations.
- If the j is too long, the trader lags behind real-time shifts, missing out on transient but profitable changes in order flow.

The optimal configuration, found to be $L = 5, j = 5$, balances stability with reactivity. This result demonstrates the importance of aligning microstructural modelling with empirical validation

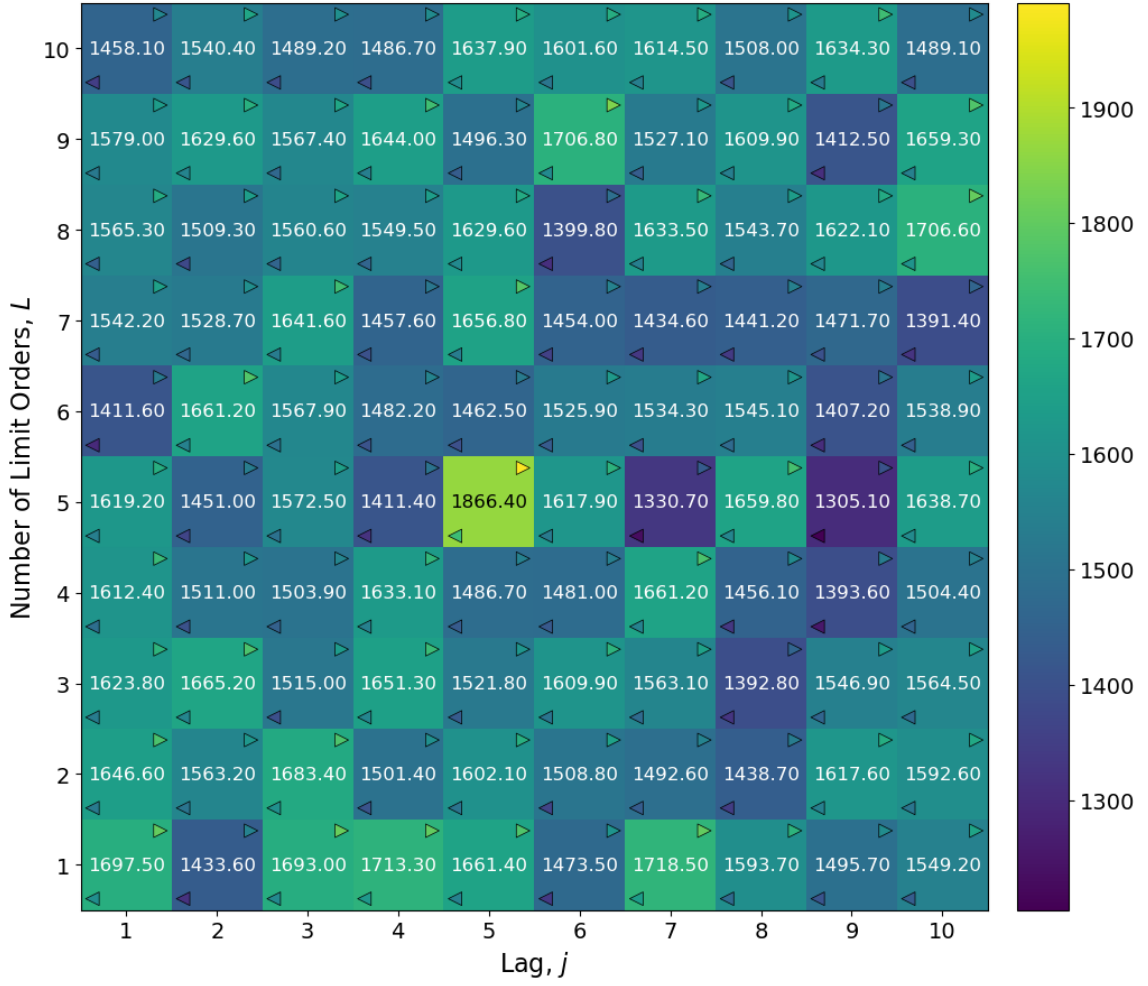


Figure 6: Heatmap of mean total profit (colour scale) across (L, j) hyperparameter pairs. Each cell's centre label gives the mean profit, and the corner triangles show ± 1 standard errors. Brighter (yellow) cells indicate higher profits. Notably, the $(5, 5)$ cell has the highest mean and the error bars do not overlap any other top-performing cells, confirming $(5, 5)$ as the optimal configuration within statistical uncertainty.

to construct high-performing trading algorithms. In essence, the MM strategy integrates principles from statistical market modelling and real-time LOB analysis with an aim of delivering consistent, risk-adjusted liquidity provisioning.

4. Results

To conduct a comprehensive evaluation of the proposed Market Maker (MM) strategy, it was implemented into the BSE framework and subjected to testing, using the baseline configuration previously specified. This statistical model, validated previously through prior KS and LB tests, reflects real-world trading environments, where order flows are often well-approximated by Poisson dynamics.

The initial experimental design involved two test conditions. Firstly, pairwise simulations between MM and each of five benchmark trader types listed in Table 1, ensuring MM could be directly compared to the other traders in isolation. Secondly, a competitive environment mimicking that of real-life markets, where all trader types coexisted within the same simulation.

In the pairwise experiments, the cumulative profit per trade for each trader was recorded over time

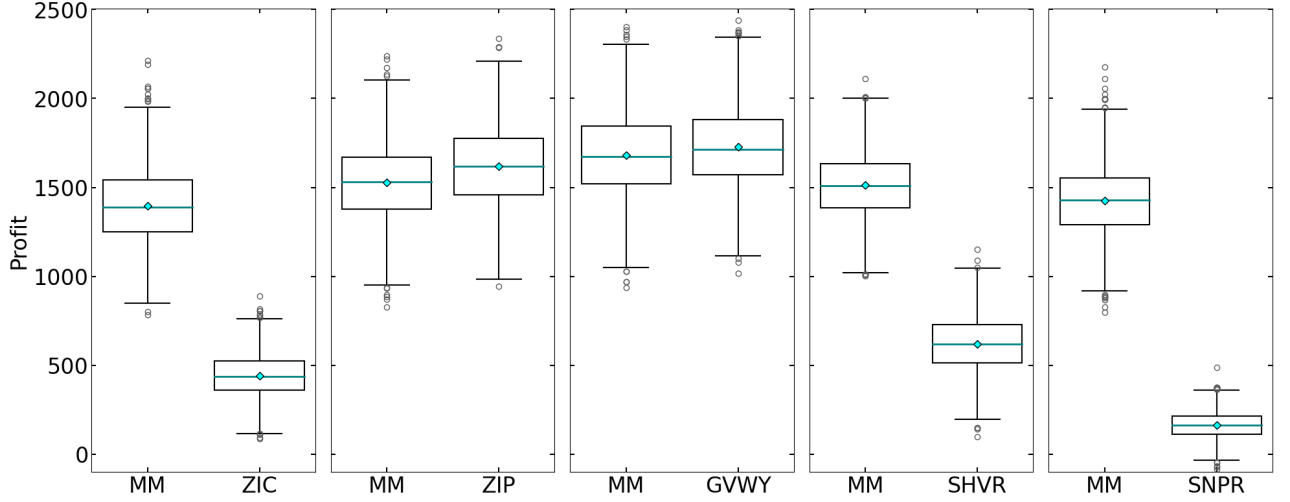


Figure 7: The final profit distributions for each pair of trading strategies, Market Maker vs ZIC, ZIP, GVWY, SHVR, and SNPR. The teal horizontal line within each box denotes the median return, while the diamond indicates the mean. Circles represent outliers. In each pairing, the Market Maker strategy exhibits consistently high median and mean returns.

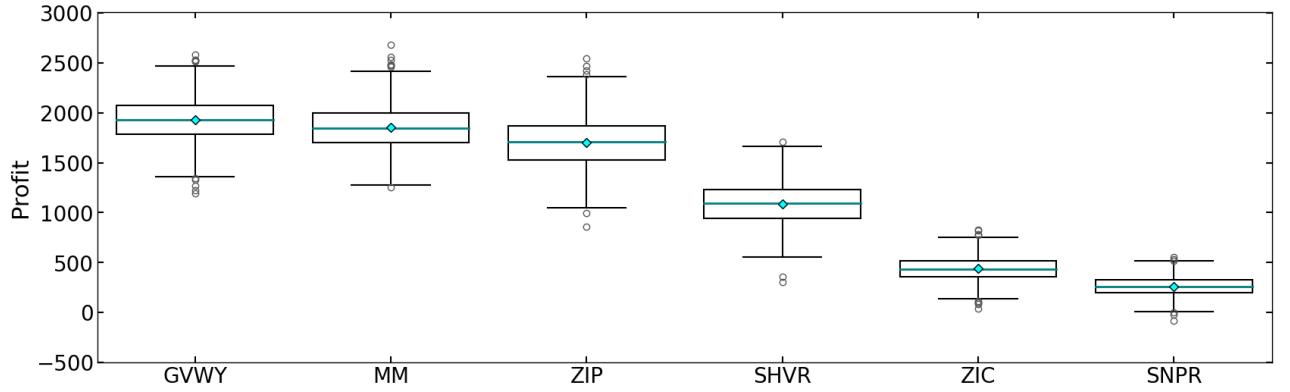


Figure 8: The final profits for the six trading strategies, GVWY, Market Maker, ZIP, SHVR, ZIC, and SNPR operating concurrently in the same simulated market. Each box shows the interquartile range, with the teal horizontal line indicating the median return and the diamond representing the mean. Whiskers capture the data spread, and outliers are shown as individual points.

and compiled into distributions of total profit, shown in Figure 7. A profit is measured against the customer's limit price: for a sell order, profit per share is the execution price minus the customer's limit price; for a buy order, it is the customer's limit price minus the execution price. In each box plot, the horizontal line marks the median, the box spans the interquartile range (IQR), circles denote outliers, and the diamond indicates the mean profit. When matched one-on-one with ZIC, SHVR, or SNPR, MM outperforms by a wide margin. Against ZIP and GVWY, however, the gap in total profit narrows and MM even falls slightly behind on average. This pattern underscores both the strength and the limitations of MM's design that anchoring quotes to a dynamically smoothed mid-price and enforcing a minimum spread delivers robust, low-volatility performance under Poisson-driven order flow, but can leave room for improvement in certain isolated scenarios.

To test robustness in a heterogeneous trading environment, all six types of agents were introduced into a shared market simulation, with results shown in Figure 8. The same three leaders from the pairwise tests (GVWY, ZIP, and MM) again dominate, underscoring the value of consistent execution,

Table 3: Key performance metrics for the six trader types in the BSE, including risk-adjusted measures (α and β) and trade-level statistics (Hit Ratio, Payoff Ratio, and Profit Factor). Each value represents the mean and its associated uncertainty across 1,000 simulation runs. Alpha and Beta have been calculated relative to Giveaway, taken as the benchmark.

Trader	α	β	Hit Ratio	Payoff Ratio	Profit Factor
GVWY	0	1	0.863 ± 0.023	0.98 ± 0.14	6.7 ± 1.8
MM	1.34 ± 0.16	-0.052 ± 0.019	0.852 ± 0.024	1.12 ± 0.15	6.2 ± 0.6
ZIP	1.21 ± 0.15	-0.027 ± 0.018	0.822 ± 0.031	1.31 ± 0.22	6.5 ± 1.8
SHVR	0.69 ± 0.15	$+0.041 \pm 0.018$	0.682 ± 0.036	1.37 ± 0.16	3.1 ± 0.7
ZIC	0.32 ± 0.08	-0.006 ± 0.009	0.793 ± 0.059	1.36 ± 0.39	5.2 ± 1.9
SNPR	0.17 ± 0.07	$+0.003 \pm 0.008$	0.714 ± 0.075	1.30 ± 0.47	4.3 ± 1.3

disciplined spread capture, and adaptive pricing. MM holds its own against GVWY, widely regarded as a high performer in simulations such as this, posting a mean return of approximately 1,847 with a moderate standard deviation of around 215. GVWY, despite being non-profit seeking in its quoting (it simply mirrors the customer’s limit price), can still earn profit by capturing the bid-ask spread on each fill, and its passive stance drives very high trade frequency. ZIP’s machine-learning driven margin adjustments keep it highly adaptive, while MM’s smoothed mid-price anchoring and enforced minimum-spread safeguard deliver steady profit with tightly controlled volatility. In contrast, SNPR and ZIC post both lower average profits and much higher volatility. Unlike GVWY, which, despite its purely reactive and random quoting, consistently captures the bid-ask spread, SNPR and ZIC lack any built-in spread-skimming mechanism, causing them to struggle in a heterogeneous, Poisson-driven market due to a lack of total executed trades.

Rather than depend on a single summary statistic, each strategy can be evaluated using a complementary suite of five metrics, covering risk-adjusted returns (alpha, α), market sensitivity (beta, β), execution quality (profit factor), and loss mitigation (hit ratio & payoff factor) [34]. Table 3 lists these metrics side by side for direct comparison. Full definitions and derivations appear in the Appendix (α and β in Appendix 8.4; profit factor in Appendix 8.5; hit ratio and payoff ratio in Appendix 8.6).

Table3 shows that, relative to GVWY as a benchmark, α (excess profit) and β (benchmark sensitivity) place MM at the top: it achieves the highest α of 1.34 ± 0.16 , indicating strong outperformance independent of market swings, and a β of -0.052 ± 0.019 , signifying minimal correlation with benchmark returns. Turning to trade level metrics, MM’s Hit ratio (fraction of profitable trades) is 0.852 ± 0.024 , meaning over 85% of its trades close in profit, on par with GVWY. MM’s payoff ratio (average profit on winning trades divided by average loss on losing trades) is 1.12 ± 0.15 , reflecting consistent, if modest, per-trade gains. Finally, MM’s profit factor (total profits over total losses) is 6.2 ± 0.6 , confirming that cumulative wins far outweigh cumulative losses. The combination of α and β with the trade-level metrics cover five orthogonal dimensions for a truly holistic performance comparison.

Although the trade-level metrics offer useful insight, a broader performance evaluation of each strategies efficiency can be evaluated by combining profit per trade with the volatility via the Sharpe ratio. The Sharpe ratio measures the profit earned per unit of risk, where risk is defined as the standard deviation of profit per trade (PPT). This provides a more comprehensive view of a strategy’s efficiency. The Sharpe ratio is calculated as

$$\text{Sharpe Ratio} = \frac{\mu_{\text{PPT}} - r_f}{\sigma_{\text{PPT}}} = \frac{\mu_{\text{PPT}}}{\sigma_{\text{PPT}}}, \quad (17)$$

where r_f is the risk-free rate, set to 0 in this study, and σ_{PPT} is the standard deviation of profit per trade for each simulation. Due to r_f being equal to 0, the Sharpe ratio effectively becomes the

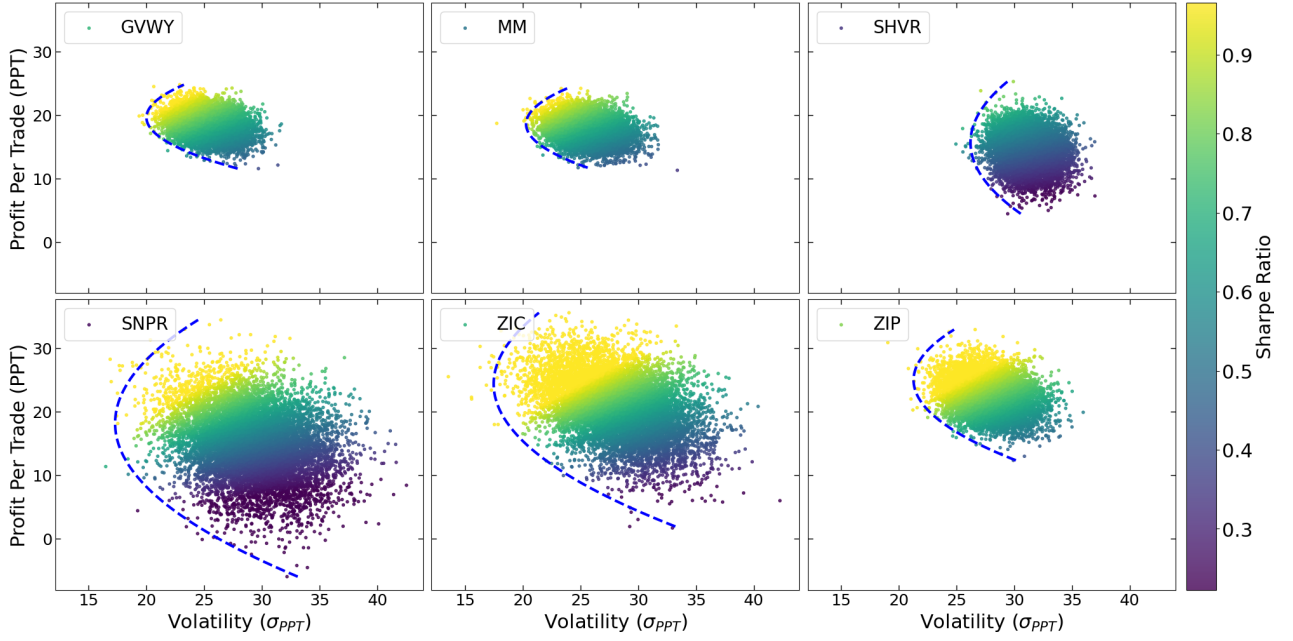


Figure 9: Comparison of the six different trading strategies, each subplot showing the distribution of profit per trade (PPT) versus volatility (risk) for that strategy, commonly referred to as the Sharpe Ratio. The points are colour-coded by Sharpe ratio, with the dashed curve approximating the efficient frontier, the best average risk–return trade-off achievable on average, by each trader.

ratio of mean profit to volatility. In traditional economic settings, the risk-free rate is equal to the central bank interest rate. A higher Sharpe ratio indicates that the strategy consistently achieves greater returns for each increment of volatility. In other words, it offers more reward per unit of risk. When viewed alongside metrics such as alpha, beta, and payoff ratio, the Sharpe ratio helps clarify whether a strategy’s profitability stems from taking on higher volatility or from genuine risk-adjusted outperformance.

The Sharpe ratio for each trader over 1,000 baseline simulations is shown in Figure 9. Each subplot shows a scatter plot of average profit per trade versus volatility per baseline simulation, with points colour-coded by Sharpe ratio. The blue dashed line represents the efficient frontier, defined as the locus of points that achieve the maximum average profit per trade for a given level of volatility. MM resides near the upper-left of its panel, close to the frontier, indicating it delivers comparatively high returns with low volatility. By contrast, SNPR, ZIC, and SHVR cluster closer to the origin, reflecting inferior risk–reward trade-offs. Moreover, MM’s tight grouping of points demonstrates its consistency: the narrow band of average return and volatility shows that it reliably maintains a favourable risk–return profile across stochastic order flows.

Taken together, the empirical evaluations demonstrate that the MM strategy provides high, consistent, and risk-adjusted returns across diverse market conditions. The findings position MM as a compelling candidate for liquidity provision strategies in real world algorithmic trading systems.

5. Discussion

A critical aspect of the investigation was the sensitivity of MM’s performance to key parameters, notably the number of ladder levels, L , and the lag, j , used in computing the effective mid-price. The optimisation experiments demonstrated that the strategy’s performance is highly contingent on the calibration settings. Specifically, while a higher L increases the number of active orders and potential market exposure, it also elevates the risk of adverse price movements. Similarly, the lag parameter

represents a trade-off between reactivity and stability. A shorter lag enhances responsiveness to market conditions at the cost of increased noise, whereas a longer lag improves stability but may render the strategy less adaptive. The identification of an optimal parameter combination ($L = 5, j = 5$), suggests that even minor deviations from these calibrated values can lead to significant variations in both the median and mean profits.

It is clear from the optimisation plot in Figure 6 that while the difference between parameter combinations in terms of total profit is often small, the impact on risk-weighted return is significant. One key reason for this lies in the early-stage dynamics of the market. Since the BSE is configured using a drip-Poisson process, order arrivals at the beginning of each session are highly stochastic. This often results in rapid price swings and heightened volatility before the market reaches a more stable state, as traders begin to converge on agreed price levels. During the volatile initial phase, the MM strategy may experience temporary drawdowns, entering negative total profit territory, before recovering in the later, more stable stages of the session. In other simulations, the strategy performs more steadily throughout. These variations highlight the sensitivity of MM to initial conditions and suggest a need for more adaptive parameter control, particularly early into the trading session.

Another important consideration is the MM wake-up period, the minimum interval between checking the LOB for updates. This represents a trade-off between computational efficiency and responsiveness. While real-world HFT systems aim for near-continuous market monitoring, the BSE imposes a logical sequencing constraint that limits how frequently updates can be processed. Similarly, while the tick size is system-defined, MM can be manually tuned to override this, quoting finer price increments and enabling the capture of smaller, high-frequency profits. These factors interact with MM's quoting logic and ultimately influence its ability to react promptly to microstructural changes.

The optimised parameters render MM competitive against other well-known strategies, as illustrated in Figure 7. Observing the distributions, MM outperforms ZIC, SHVR, and SNPR in terms of both median and mean profit. Here, MM's box typically sits higher on the profit scale, reflecting fewer low-profit runs and more frequent moderate-to-high runs. Despite MM's consistent performance, both GVWY and ZIP occasionally outperform it in terms of median and mean profit. GVWY, with its straightforward approach of mirroring customer order prices, tends to capitalise on high trade frequencies with profit simply being the spread between the customer's price and the best opposing quote on the book. This leads to superior cumulative profits when the environment is particularly favourable. Similarly, ZIP's adaptive learning mechanism allows dynamic pricing adjustment, enabling exploitation of transient opportunities more aggressively than MM. Notably, these advantages become especially pronounced when the market is less complex, for instance, when only two traders are present. In such scenarios, market dynamics are simpler and less volatile, giving strategies like ZIP and GVWY a clearer edge. The reduced competition and lower volatility allow these strategies to achieve higher peak profits, even though they might come at the cost of increased variability compared to MM's more balanced, risk-adjusted approach.

While MM's IQR isn't the tightest, the strategy still delivers a consistently profitable baseline. In practice, this means MM rarely suffers large losses or deep drawdowns, giving it a strong risk-adjusted profile. A narrow IQR often signals uniform results, but MM's slightly wider spread actually highlights its adaptability: it performs steadily across varied market conditions without extreme swings. Occasionally, MM will register outliers, typically when its quoting lag prevents it from capturing the very short-lived volatility at market open. Likewise, in thin markets with few participants, matched trades occur less often, which reduces the chances of earning the bid-ask spread and therefore lowers overall profits.

Although one-to-one comparisons are useful for assessing pairwise performance, they can oversimplify market volatility and fail to capture the complexity of multi-agent dynamics. Figure 8 highlights a more realistic, competitive environment. The plot clearly reveals a ranking of the strategies in terms of mean profit when run simultaneously. In the highest tier, GVWY, MM, and ZIP consistently achieve

larger profits, with distributions around or above 1500 – 2000 in central tendency. In contrast, SHVR, ZIC, and SNPR fall into a lower range, typically ranging from a few hundred to just over 1000 in profit. This underscores that certain strategies, particularly those that actively adapt or maintain a robust quoting system, tend to fare better in a multi-agent environment, while they get suppressed in 1-1 markets. This adds emphasis on the parameter choice, highlighting that a dynamic lag over the course of the market is a point of expansion within the strategy.

Closer inspection of the box for GVWY indicates some extreme positive outliers exceeding 2500, suggesting that in certain scenarios, the straightforward approach of placing limit orders very close to the customer’s limit price pays off well. On the downside, GVWY displays a number of relatively low-profit outliers, reflecting that while it can excel under stable or predictable flow conditions, it remains vulnerable to rapidly shifting environments due to the lack of adaptability. MM occupies a similarly high return range, with a distribution around 1700 and 2000. The narrower IQR reflects a high degree of consistency, with fewer extreme negative outliers compared to GVWY. This reinforces the notion that systematically quoting around a smoothed mid-price, in a market populated by diverse strategies, enables MM to lock in relatively stable profiting sessions. ZIP appears closely behind the two aforementioned strategies, with a plot that overlaps significantly in terms of performance, though exhibiting a slightly broader spread. The presence of outliers above 2400 suggests that ZIP is capable of capitalising on volatile market conditions. However, the distribution also dips further below MM’s, revealing adverse scenarios where ZIP’s adaptive margin fails to keep pace with rapid shifts in the limit order book, resulting in numerous outliers below.

Moving further right, SHVR’s distribution, while still centred above 1000–1200, rests clearly below the top three strategies. SHVR’s median and mean cluster in the mid-range, with some outliers topping 1600, but also a handful of negative results dipping below 500. This pattern reflects SHVR’s strategy of shaving off the best bid and ask prices. While this strategy can secure profits in many stable intervals, it lacks the robust adaptive mechanisms that protect it from larger swings or capital-intensive opportunities. ZIC and SNPR round out the lower end of profitability in the six-agent environment. ZIC’s box is centred around 500, with some simulations achieving moderate gains of 800–900, but also negative outliers around 250. SNPR, as in the pairwise plots, demonstrates the worst performance overall, with the most results below 500 in return, and runs even dipping into negative territory. SNPR’s time-sensitive sniping approach appears less effective when pitted against multiple adaptive or stable-quoting agents, resulting in comparatively lower median outcomes. This is mainly due to SNPR becoming active in the later stages of the market, resulting in a much smaller amount of total trades taking place per session.

The MM distribution highlights a certain stability in the six-agent setup, with a number of positive outliers. These outliers represent simulations where the MM’s two-sided ladder quoting mechanism lined up especially well with transient price and liquidity imbalances, enabling it to realise significant gains. Such scenarios arise when the MM quickly updates its quotes to capture newly available liquidity spikes or to avoid sudden adverse selection, illustrating the strategy’s capacity for opportunistic profits under favourable, albeit stochastic, market conditions. Conversely, MM exhibits relatively fewer negative outliers compared to many of the competing strategies. Although present, indicating short-lived episodes of unprofitability, such outlier events appear less numerous and less extreme than those encountered by the likes of GVWY, ZIP and SHVR. The reduced downside risk demonstrates the MM’s resilience, should the market suddenly shift outside the quoting strategy’s intended range, or if volatility spikes beyond expectations, MM typically recovers quickly by readjusting quotes around the calculated mid-price. Thus, even where the MM suffers losses, the bounded nature of these negative runs prevents any crippling drawdowns.

Comparing all six traders side by side highlights the interplay between agent strategies and the emergent market conditions. GVWY’s simpler approach can flourish under moderate competition, posting consistently high profits and notable positive outliers, but can also suffer larger losses in

turbulent sessions. GVWY would likely be the worst performer if transaction costs were introduced, since it skims the spread on every trade and has no recollection for a minimum profit threshold. MM’s ladder-based quoting excels at maintaining a stable profit stream through its customisable profit margin and avoids the biggest drawdowns. Conversely, purely reactive strategies such as SNPR appear disadvantaged when multiple adaptive or moderate-risk strategies populate the market.

In real financial markets, HFT’s need to balance generating profits with managing risk, a trade-off that can be evaluated using performance metrics such as those in Table 3. When evaluating risk-adjusted returns using alpha and beta, SNPR and ZIC exhibit small positive alpha values, ranging from 0.17 ± 0.07 to 0.32 ± 0.08 , alongside near-zero or slightly negative beta values. This indicates that although these strategies occasionally outperform GVWY, their excess returns are limited and not strongly correlated with market movements. This behaviour stems from their underlying logic; ZIC selects prices randomly within a constrained range, making it largely indifferent to market conditions, while SNPR remains inactive for much of the session, only becoming aggressive near the end. As a result, both strategies behave more idiosyncratically and reactively, leading to lower correlations with GVWY’s performance and reduced consistency in their returns. In contrast, SHVR shows a higher alpha of 0.69 ± 0.15 and a small positive beta of 0.041 ± 0.018 , indicating that it moderately outperforms GVWY while still loosely tracking the benchmark’s behaviour. This reflects SHVR’s design, which involves undercutting the best bid or ask by small increments. While not adaptive, this tactic allows it to remain competitive in the book and benefit from frequent executions, leading to modest but steady gains.

The standout risk-adjusted performers are ZIP and MM, exhibiting the highest alpha values of 1.21 ± 0.15 and 1.34 ± 0.16 respectively, coupled with negative beta values. Here, the higher alpha does not imply that these strategies always achieve the highest absolute profits; rather, it indicates that they consistently deliver returns above the benchmark regardless of how GVWY performs. ZIP achieves this through its machine-learning inspired adaptive pricing, which adjusts margins based on recent market feedback. MM, on the other hand, anchors its quotes around a smoothed mid-price and enforces a minimum profit spread, which cushions it during volatile phases. Their negative beta values suggest that both strategies often perform well when GVWY underperforms, essentially providing an edge that is independent of the benchmark’s movements. In this context, alpha captures their structural advantage, while beta reflects their inverse exposure to GVWY’s random quoting behaviour.

Next, consider trade quality as measured by hit ratio. GVWY’s hit ratio of 0.863 ± 0.023 means that approximately 86.3% of its trades are profitable, an impressively high number. This is largely driven by its passive nature of quoting at the customer’s limit price, often leading to quick fills. Most other strategies fall in the 68%–85% range, reflecting different levels of risk and aggressiveness in trade selection. For instance, SHVR’s hit ratio of 0.682 ± 0.036 suggests it wins less than 70% of its trades, consistent with its slightly riskier approach of undercutting the best quotes. ZIP and MM register hit ratios of 0.822 ± 0.031 and 0.852 ± 0.024 , respectively, putting them close to GVWY. MM’s high hit ratio is largely a result of its systematic two-sided quoting mechanism, which maintains a persistent presence in the LOB across multiple levels around a smoothed mid-price. This increases the likelihood of MM’s orders being matched by incoming customer orders and filled, especially during periods of high liquidity. Furthermore, the use of a profit margin buffer and effective mid-price ensures that most of these fills are at favourable prices, contributing to the strategy’s ability to close the majority of trades in profit. Unlike more reactive or opportunistic strategies, MM’s structured and passive nature naturally leads to a higher proportion of winning trades, albeit with smaller per-trade gains, an outcome typical of liquidity-providing strategies that prioritise execution consistency over speculative positioning [35].

The payoff ratio quantifies the average profit per winning trade relative to the average loss per losing trade. SHVR achieves the highest payoff ratio of 1.37 ± 0.16 , meaning each win yields about 37%

more than each loss, though its lower hit ratio limits overall profitability. ZIC, with a payoff ratio of 1.36 ± 0.39 , achieves moderate net profit. In contrast, the MM strategy records a lower payoff ratio of 1.12 ± 0.15 , indicating its losses are slightly larger than its gains on a per-trade basis. This reflects MM’s systematic quoting behaviour, where it frequently captures small spreads in stable conditions, but occasionally experiences losses when prices shift sharply before it can re-quote. However, this lower per-trade payoff is more than offset by MM’s high hit ratio of 85.2%, demonstrating that its consistency in closing trades at a profit outweighs the magnitude of occasional losses. This once again is characteristic of liquidity providing strategies, which often sacrifice high individual gains in favour of frequent, small profits [19].

To measure overall profitability, the profit factor measures total gross profits versus total gross losses, with values above 1 indicating profitability and above 2 suggesting strong performance. GVWY stands the highest at 6.7 ± 1.8 , reflecting its high hit ratio and simpler strategy of matching orders at the limit price. ZIP follows at 6.5 ± 1.8 , boosted by adaptive margin tuning. Meanwhile, MM achieves 6.2 ± 0.6 , indicating a robust risk-reward balance and disciplined quoting, though its nominally lower profit factor still falls within the overlapping uncertainties of GVWY and ZIP. Hence, it is not possible to conclude that GVWY or ZIP definitively outperform MM in terms of overall profitability. ZIC, at 5.2 ± 1.9 , performs respectably, but lacks the consistency seen in MM’s design. In contrast, SHVR at 3.1 ± 0.7 and SNPR at 4.3 ± 1.3 appear more vulnerable to adverse price movements.

Following the trade-level analysis, a broader perspective is taken using the efficient frontier framework, shown as the blue dashed line in Figure 9. Here, the horizontal axis measures volatility (the standard deviation of per-trade profit) and the vertical axis plots mean profit per trade. The efficient frontier thus traces the maximum attainable profit for any given level of risk, while the points are coloured to demonstrate the Sharpe ratios. In this context, the efficient frontier is just the average highest envelope of risk–return trade-off for each strategy. While each strategy produces its own elliptical cluster of points, key distinctions emerge. SNPR, for example, spreads from roughly 20 to 40 in volatility and delivers highly variable Sharpe ratios, an artifact of its reactive, time-sensitive behaviour. In contrast, GVWY and ZIP occupy tighter volatility bands with moderate mean returns: GVWY by virtue of sheer trade frequency alone, ZIP through its adaptive margin adjustments. ZIP can touch efficient frontier “peaks” under certain simulated conditions, but its sessions often drift off the frontier when noisy price signals dominate.

MM, however, consistently nests along the frontier’s upper segment, combining moderate volatility with some of the highest mean profits per trade. Its cluster of points corresponds to superior Sharpe ratios, reflecting regular gains for each unit of risk taken, shown by little variation in colour and a flatter frontier curve. Unlike GVWY, whose wins depend on random proximity to incoming customer prices, MM’s successes are intentional and repeatable. By anchoring bids and asks around a smoothed mid-price and enforcing a calibrated spread, MM avoids over-exposure in sudden market swings and captures reliable profits in calmer intervals. Although ZIP occasionally matches MM’s Sharpe highs, it rarely sustains them; MM’s disciplined quoting and predictable fill probabilities ensure profitability without forecasting, making it robust across Poisson-driven order flows. Thus, even if multiple strategies map similar efficient frontier shapes, MM’s overall advantage becomes clearer when considering absolute profitability and stability. Its design enables a robust, high-frequency response to order flow while mitigating downside risk, traits that are essential in both simulated and real trading environments. By balancing stable spread capture with adaptive mid-price anchoring, MM maintains superior Sharpe ratios and outperforms peers not just in shape, but in substance.

A further insight is that visually similar efficient frontiers do not guarantee equivalent profit outcomes. While these curves effectively summarise how well a strategy converts risk into return, they often obscure key structural differences, such as execution frequency, inventory constraints, or exposure to adverse selection. In Poisson-based simulations, MM benefits from a consistent stream of matched orders, thanks to its two-sided quoting and anchored placement around the smoothed mid-price.

This stability increases the likelihood of profitable fills, even without predictive modelling, ultimately elevating cumulative returns beyond what the efficient frontier alone might suggest.

Finally, while the BSE simulation framework effectively highlights structural differences between strategies, it operates under idealised assumptions of zero transaction costs and a simplified market microstructure. Still, the evidence presented underscores that a systematic, mid-price-anchored market-making strategy can deliver strong, risk-adjusted returns under stochastic order flow, offering resilience against volatility and capitalising on short-term inefficiencies.

In general, MM and ZIP stand out for their high alpha, consistently earning returns that are not closely tied to the performance of GVWY. MM may not claim the top spot in every statistic, its payoff ratio and total cumulative profit, for example, fall short of some alternative strategies, but it makes up for this with exceptionally reliable execution (high hit rate) and efficient profit capture (strong profit factor). In contrast, SHVR and ZIC show much more variability in their results, and SNPR generally underperforms on most metrics. This comparison makes it clear that no single measure tells the whole story. Only by looking at a combination of hit rate, payoff ratio, profit factor, and alpha can it be shown how well each strategy balances risk, reward, and adaptability. MM’s real advantage is its well-rounded profile: not the absolute leader in any one category, but steadily profitable and repeatable across all of them.

6. Conclusion

This report has proposed a ladder-based Market-Making (MM) strategy within the Bristol Stock Exchange, illustrating how systematic quoting around a smoothed mid-price can deliver robust, risk-adjusted profitability under Poisson-driven market dynamics. While other strategies occasionally match or exceed MM’s performance on specific metrics or under isolated conditions, MM consistently outperforms across key dimensions, most notably mean per-trade profits, alpha, and profit factor, demonstrating its resilience and reliability in diverse trading environments.

The tuning of ladder depth and mid-price lag emerges as critical to achieving an optimal balance between responsiveness and stability. The agent-based framework used in this study allows for controlled exploration of such parameters and helps isolate the effects of microstructural assumptions. However, the simulation environment also introduces some limitations. Chief among these are the absence of transaction costs and slippage. Furthermore, while Poisson processes offer analytical tractability, they may fall short in capturing the temporal clustering and heterogeneous behaviours commonly seen in real financial markets.

To strengthen the practical relevance of this work, there are several avenues for future development. Firstly, incorporating more realistic cost models, including fees per-trade, exchange rebates, and slippage, would help assess the net profitability of the MM strategy in real-world conditions. Such enhancements would also highlight the comparative advantage of structured, adaptive strategies like MM over simpler or random approaches such as GVWY, which are more susceptible to cost-related erosion.

Secondly, introducing adaptive parameter calibration would improve the MM strategy’s resilience in volatile or rapidly evolving markets. Currently, MM’s performance depends on fixed values for parameters such as ladder depth and lag. Implementing mechanisms to dynamically adjust these settings based on real-time market signals, such as shifts in order flow or volatility, could significantly enhance both responsiveness and profitability.

Finally, extending the simulation framework to support a more diverse set of agent behaviours would provide a richer and more rigorous stress test of the MM strategy. In live markets, traders employ a wide array of tactics, many of which deviate from a Poisson assumption. Incorporating agents that mimic realistic, data-driven strategies would allow for a more comprehensive evaluation of how MM performs under competitive and heterogeneous market conditions, while also uncovering potential

edge cases or emergent inefficiencies.

Overall, our results show that by harnessing microstructure signals, enforcing disciplined ladder-based quoting, and empirically tuning key parameters, the MM strategy offers a powerful framework for liquidity provision in high-frequency trading. Although further refinement, such as more realistic cost models and adaptive parameter updates, will be needed to translate this approach into live markets, this work lays a firm foundation for building robust, adaptive market-making algorithms.

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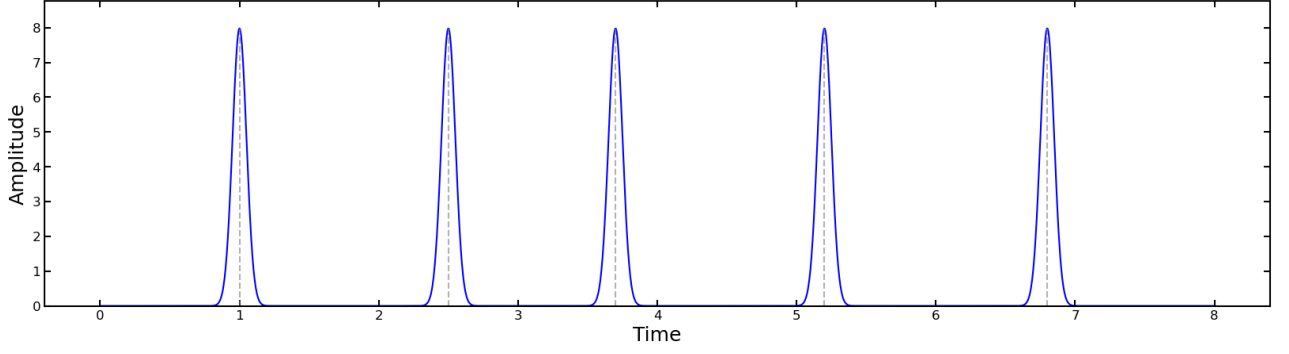


Figure 10: Approximation of Dirac-delta spikes at discrete arrival times, τ_K . Each tall, narrow blue curve is a Gaussian kernel of width σ centred at an event time, summing to approximate $\sum_K \delta(t - \tau_K)$. The vertical dashed lines mark the true arrival instants.

8. Appendices

8.1. Visualisation of the Dirac Delta Representation

In continuous time point processes, a sequence of event times, τ_K , can be represented as a counting function

$$N(t) = \sum_K 1_{\tau_K \leq t}$$

which jumps by one at each arrival τ_K . Formally, the derivative can be written as

$$\frac{dN}{dt} = \sum_K \delta_1(t - \tau_K)$$

where $\delta(t)$ is the Dirac delta function. Although δ is not a true function but rather a distribution, it has three key properties

- (i) Zero everywhere except at the origin: $\delta(t) = 0$ for $t \neq 0$.
- (ii) Unit integral: $\int_{-\infty}^{\infty} \delta(t) dt = 1$.
- (iii) Sifting property: For any smooth f , $\int f(t) \delta(t - \tau_K) dt = f(\tau_K)$

It is not possible to plot $\delta(t)$, so in practice it is common to approximate it by a very narrow Gaussian of width σ :

$$\delta_\sigma(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right),$$

which converges to $\delta(t)$ as $\sigma \rightarrow 0$. This is illustrated in Figure 10, where a few example order arrival times were used to build a fine time grid, and then sum up the narrow Gaussians centred at each τ_K . When plotted, each spike appears as a tall, narrow bump, and dashed vertical lines mark the exact event times. As σ is made smaller, these bumps become ever sharper, visually mimicking the idealised Dirac spikes.

8.2. Baseline Setup

To ensure that the simulation results are statistically robust and to justify the choice of 1000 simulation trials, a convergence analysis based on Monte Carlo error theory was performed, demonstrated in Figure 11. In Monte Carlo simulations, key outcome metrics (in this case, the end-of-trial profit for each trader type) are estimated by averaging the results over many independent simulation runs. The theoretical basis for this approach is that the standard error (SE) of the mean

decreases proportionally to $1/\sqrt{N}$, where N is the number of trials. This relationship implies that as more trials are run, the uncertainty in the estimated mean profit should diminish, converging to a stable value. The standard error is calculated as

$$SE = \frac{\sigma}{\sqrt{N}}, \quad (18)$$

where σ is the sample standard deviation computed from the first N trials.

Based on the convergence analysis, it can be concluded that 1000 simulation trials provide a statistically sufficient sample size, ensuring that the estimated performance metrics (i.e., the mean trader profits) are both stable and reliable. Consequently, 1000, 120s long simulations with a drip-Poisson order flow and 0.1s order interval is taken to be the 'baseline' setup. The short interval of 0.1s enables the market to operate in high speed, improving comparability to real financial markets.

8.3. Mean return and standard deviation of return from Figure 8

Table 4: Mean return and corresponding standard deviation (σ) for each trader type. The results highlight both the central tendency of returns and the variability experienced by each strategy in the BSE environment.

Trader	Mean Return	σ Avg Return
GVWY	1936.93	224.73
MM	1847.16	215.34
ZIP	1686.30	247.76
SHVR	1079.75	198.28
ZIC	443.01	114.47
SNPR	252.73	92.84

8.4. Giveaway as a benchmark for Alpha and Beta

To assess the performance of each trading strategy relative to the GVWY benchmark, we employ a regression framework inspired by the Capital Asset Pricing Model (CAPM). For a given strategy i , the relationship is modelled between its return, $R_{i,t}$ and the return of the GVWY benchmark R_{GVWY} as follows

$$R_{i,t} = \alpha + \beta_i R_{GVWY,t} + \epsilon_{i,t}$$

where $R_{i,t}$ is the return of strategy i at time t , $R_{GVWY,t}$ is the return of GVWY at time t , α_i is the intercept, representing the average excess return of strategy i relative to the benchmark, β_i measures the sensitivity of strategy i 's return to changes in the benchmark return, and $\epsilon_{i,t}$ is the error term.

When GVWY is regressed against itself, obtained is

$$R_{GVWY,t} = 0 + 1 \cdot R_{GVWY,t} + 0$$

implying that $\alpha_{GVWY} = 0$ and $\beta_{GVWY} = 1$. This normalisation serves as the reference point.

For the other strategies, the interpretation of results is as follows:

- A positive α_i indicates strategy i consistently earns returns in excess of the GVWY benchmark, while a negative α_i indicates underperformance.

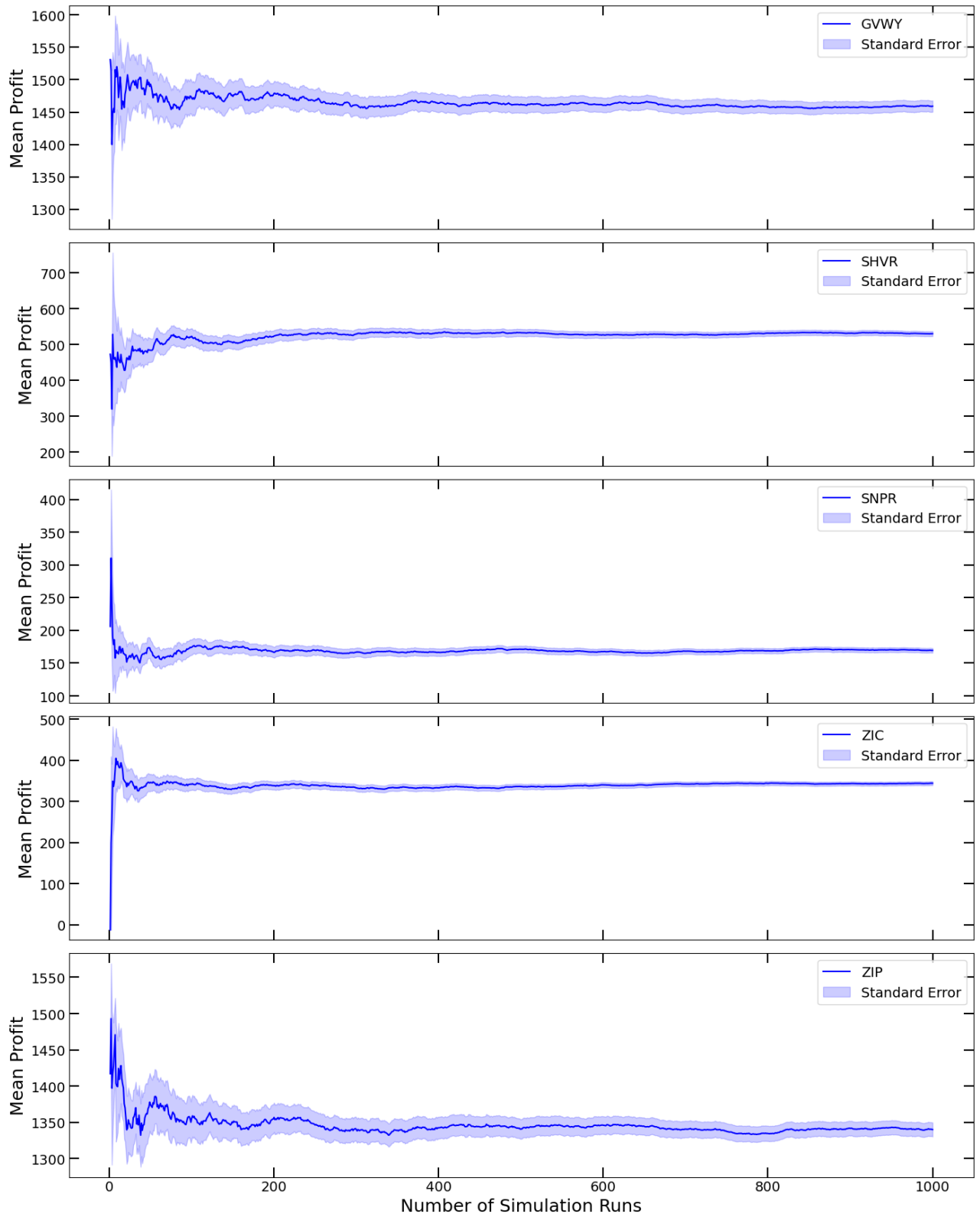


Figure 11: Cumulative running mean of each trader's end of trial profit over an increasing number of simulation runs, 1 to 1000 for simulations lasting 120s. The shaded area around each line indicates the standard error of the mean. As the number of runs increases, the curves converge and the uncertainty bands narrow, indicating that around 1000 runs are sufficient for stable, reliable estimates of trader profitability.

- The β_i coefficient captures how strongly the strategies returns respond to the benchmarks movements. A $\beta_i > 1$ suggests that the strategy is more volatile than the benchmark, whereas $\beta_i < 1$ indicates lower sensitivity. A negative β_i would imply that the strategy tends to move inversely to the benchmark.

The parameters α_i and β_i are estimated using Ordinary Least Squares (OLS) regression on a time series of returns. This method minimises the sum of squared errors, providing the best linear unbiased estimators for the parameters under standard assumptions.

In summary, this CAPM-based regression allows us to decompose each strategy's performance into:

- α_i : The excess return not explained by market movements.
- β_i : The measure of systematic risk or sensitivity relative to the GVWY benchmark.

8.5. Profit Factor

Every time a trade is executed, it is recorded as either a loss or gain, with the amount for that trader. The profit factor can then be calculated as

$$\text{Profit Factor} = \frac{\text{Total Profit}}{|\text{Total Loss}|}$$

A profit factor of 2 means that over the simulation, the total profits were twice the total losses. A profit factor of one would mean they are the same.

8.6. Hit Ratio and Payoff Ratio

The hit ratio measure the fraction of trades that are profitable while the payoff ratio is the ratio of average profit on winning trades to average loss on losing trades. The Hit ratio can be calculated as

$$\text{Hit Ratio} = \frac{\text{Number of Winning Trades}}{\text{Total Number of Trades}}$$

while the payoff ratio can be calculated as

$$\text{Payoff Ratio} = \frac{\text{Average Profit on Winning Trades}}{|\text{Average Loss on Losing Trades}|}$$

A high hit ratio but low payoff ratio might indicate many small wins but a few large losses. A lower hit ratio with a high payoff ratio suggests that although a strategy loses more often, the wins significantly outweigh the losses in magnitude.

8.7. Market Maker Code

```
class MarketMaker:
    def __init__(self, params):
        # Parameters with defaults
        # minimum price increment
        self.tick_size = params.get("tick_size", 1)
        # maximum absolute position allowed
        self.inventory_limit = params.get("inventory_limit", 200)
        # seconds between re-quotes
        self.requote_interval = params.get("requote_interval", 0.02)
        # order size
        self.order_quantity = params.get("order_quantity", 5)
        # number of price levels per side
```

```

self.levels = params.get("levels", 5)

# Ignore balance and only track inventory
self.position = 0
self.last_quote_time = None
# Instead of single orders, we use lists for active orders
self.active_buy_orders = []
self.active_sell_orders = []
# Store midprices to optionally use an older value if desired
self.midprice_history = []

def update_lob(self, lob_data):
    """
    Expects lob_data as a dictionary with keys:
        "BestBid" and "BestAsk"
    """
    best_bid = public_data['bids']['best']
    best_ask = public_data['asks']['best']

    #best_bid = lob_data.get("BestBid", TBSE_SYS_MIN_PRICE)
    #best_ask = lob_data.get("BestAsk", TBSE_SYS_MAX_PRICE)
    return best_bid, best_ask

def get_midprice(self, best_bid, best_ask):
    """
    Calculate the midprice from the current best bid and ask.
    If both are available, return their average; otherwise
    return the available price.
    """
    if best_bid is not None and best_ask is not None:
        return (best_bid + best_ask) / 2.0
    elif best_bid is not None:
        return best_bid
    elif best_ask is not None:
        return best_ask
    else:
        return (TBSE_SYS_MIN_PRICE + TBSE_SYS_MAX_PRICE) / 2.0

def cancel_order(self, order):
    """
    Simulate order cancellation by marking the order inactive.
    """
    order["active"] = False
    print(f"Cancelled order: {order}")

def place_order(self, side, price, quantity):
    """
    Simulate placing an order. Returns an order dictionary.
    """
    order = {"side": side,
            "price": price,
            "quantity": quantity,

```

```

        "active": True}
print(f"Placed order: {order}")
return order

def re_quote(self, lob_data, current_time):
    best_bid, best_ask = self.update_lob(lob_data)
    midprice = self.get_midprice(best_bid, best_ask)
    self.midprice_history.append(midprice)
    # Optionally, use an older midprice
    if len(self.midprice_history) >= 3:
        effective_mid = self.midprice_history[-3]
    else:
        effective_mid = midprice

    # Build a ladder of desired buy prices.
    buy_prices = []
    for level in range(self.levels):
        # Start with best_bid + tick_size (most aggressive) and then decrement.
        price = best_bid + self.tick_size - level * self.tick_size
        # To ensure we never buy above fair value, clip to midprice.
        price = min(price, midprice - self.tick_size)
        buy_prices.append(int(price))

    # Build a ladder of desired sell prices.
    sell_prices = []
    for level in range(self.levels):
        # Start with best_ask - tick_size and then increment.
        price = best_ask - self.tick_size + level * self.tick_size
        # To ensure we never sell below fair value, clip to midprice.
        price = max(price, midprice + self.tick_size)
        sell_prices.append(int(price))

    # Safety: if buy ladder overlaps sell ladder, re-adjust around effective_mid.
    if max(buy_prices) >= min(sell_prices):
        buy_prices = [int(effective_mid - self.tick_size - i * self.tick_size)
                       for i in range(self.levels)]
        sell_prices = [int(effective_mid + self.tick_size + i * self.tick_size)
                       for i in range(self.levels)]

    # Cancel any active orders not matching the desired ladder.
    for order in self.active_buy_orders[:]:
        if order["price"] not in buy_prices:
            self.cancel_order(order)
            self.active_buy_orders.remove(order)
    for order in self.active_sell_orders[:]:
        if order["price"] not in sell_prices:
            self.cancel_order(order)
            self.active_sell_orders.remove(order)

    # Place new buy orders for missing price levels (if within inventory limits).
    current_buy_prices = [order["price"] for order in self.active_buy_orders]
    for price in buy_prices:

```

```

        if price not in current_buy_prices and self.position > -self.inventory_limit:
            new_order = self.place_order("BUY", price, self.order_quantity)
            self.active_buy_orders.append(new_order)

# Place new sell orders for missing price levels.
current_sell_prices = [order["price"] for order in self.active_sell_orders]
for price in sell_prices:
    if price not in current_sell_prices and self.position < self.inventory_limit:
        new_order = self.place_order("SELL", price, self.order_quantity)
        self.active_sell_orders.append(new_order)

self.last_quote_time = current_time

def on_fill(self, order, fill_price):
    """
    When an order is filled, update the maker's inventory.
    """
    if order["side"] == "BUY":
        self.position += order["quantity"]
    elif order["side"] == "SELL":
        self.position -= order["quantity"]
    order["active"] = False
    print(f"Order filled: {order} at {fill_price}. New position: {self.position}")

def run(self, lob_stream, market_order_hits):
    """
    Runs the market maker over a stream of LOB events.
    :param lob_stream: Iterable of events (dicts) with keys "Time" and "LOB".
    :param market_order_hits: Function taking (order, lob_data) and returning
    True if order is filled.
    """
    for event in lob_stream:
        current_time = event["Time"]
        lob_data = event["LOB"]

        # Re-quote if enough time has passed.
        if (self.last_quote_time is None) or
            (current_time - self.last_quote_time >= self.requote_interval):
            self.re_quote(lob_data, current_time)

        # Check for fills on active orders.
        for order in self.active_buy_orders[:]:
            if market_order_hits(order, lob_data):
                self.on_fill(order, lob_data.get("BestAsk"))
                self.active_buy_orders.remove(order)
        for order in self.active_sell_orders[:]:
            if market_order_hits(order, lob_data):
                self.on_fill(order, lob_data.get("BestBid"))
                self.active_sell_orders.remove(order)

```

END OF REPORT